Set Operations

Five ways to combine two sets — and every one of them is a connective you already know.

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$A \cup B$

Today

AGENDA

01 Union and intersection

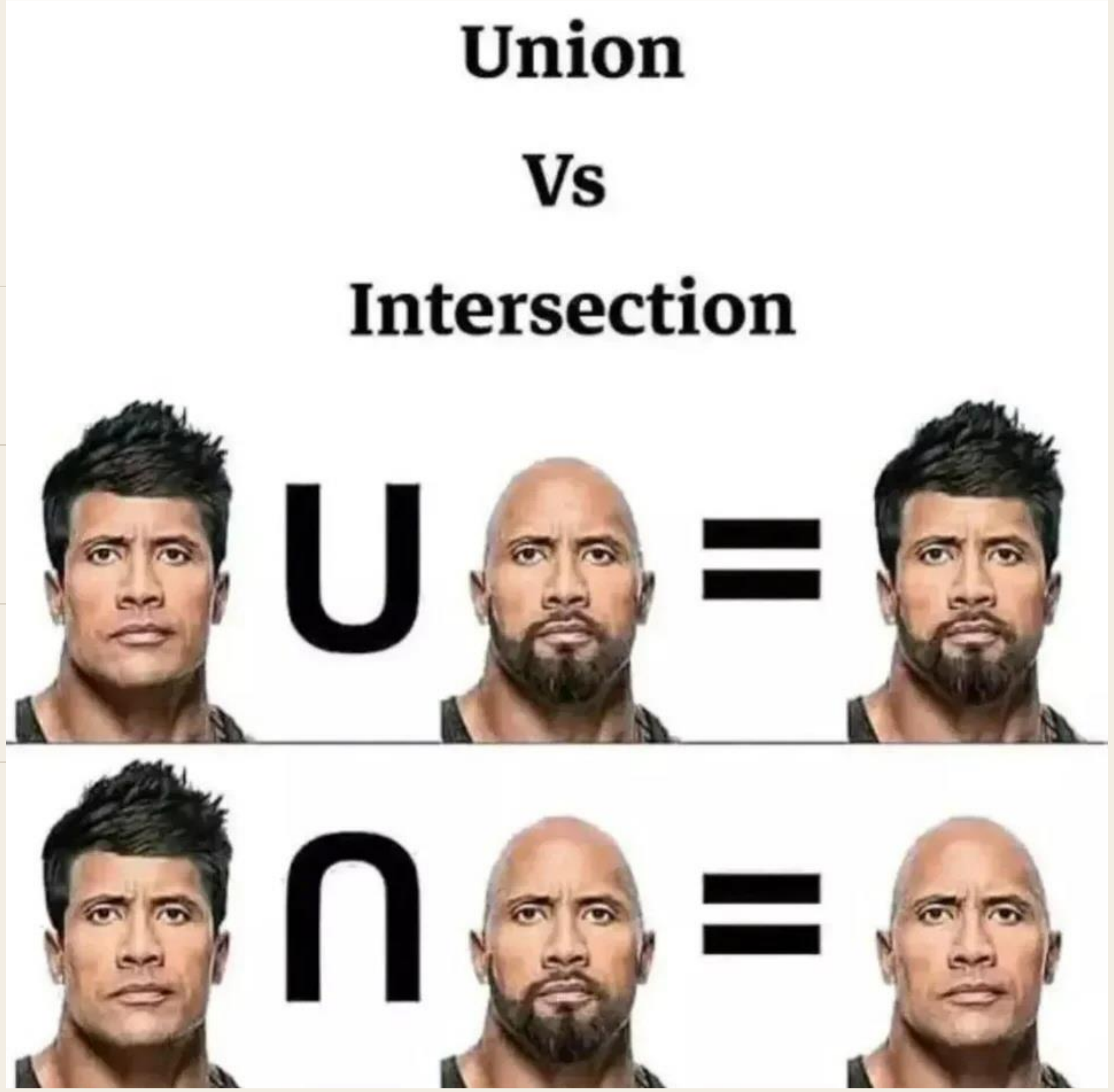
02 Counting a union without double-counting

03 Difference, complement, symmetric difference

04 Reading regions off a three-circle diagram

05 The identities, and three ways to prove one

06 Functions, and how bijections measure size

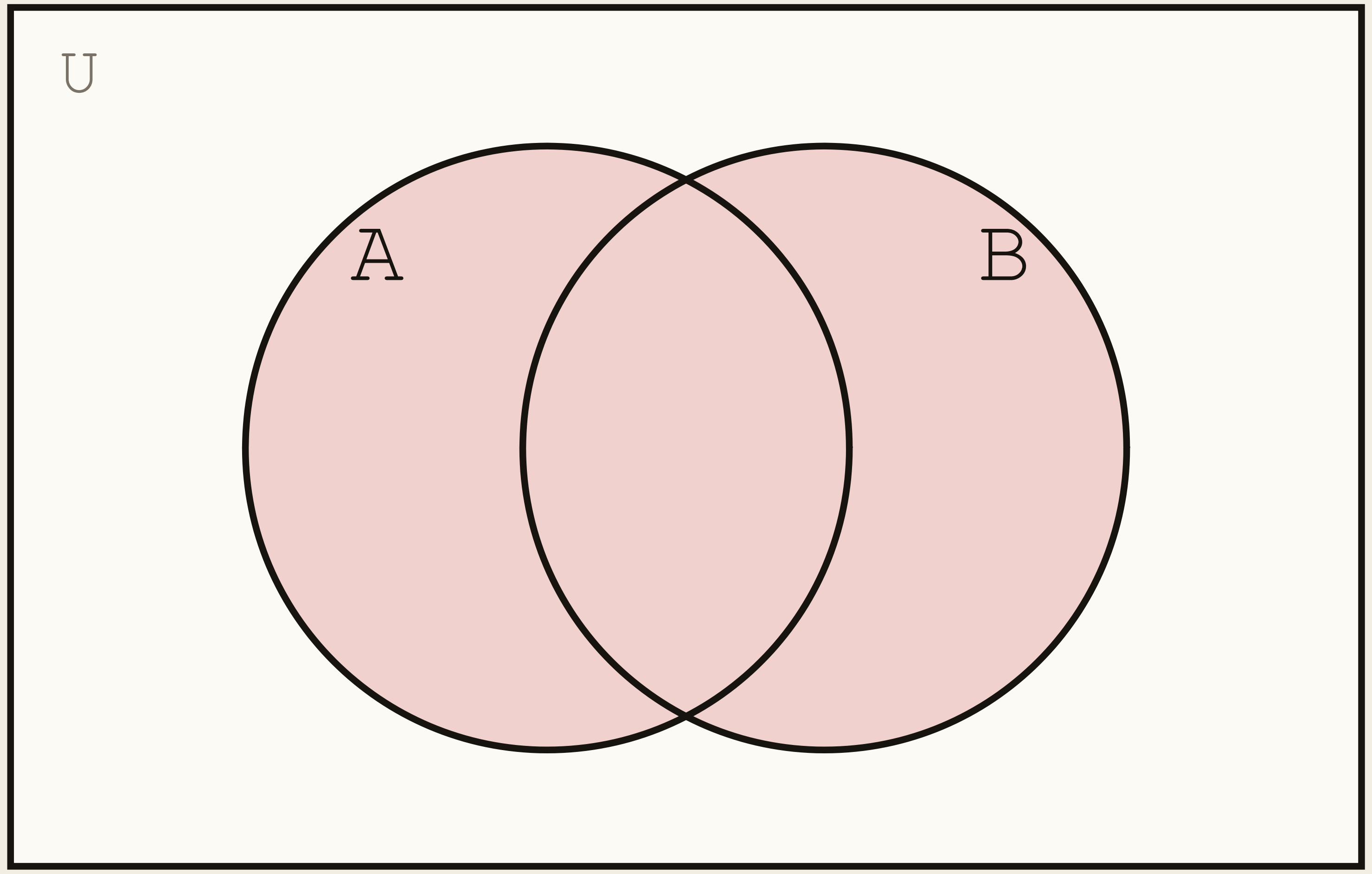


Union

DEFINITION

The **union** of A and B, written $A \cup B$, contains those elements that are in A, or in B, or in both.

$$A \cup B = \{x \mid x \in A \lor x \in B\}$$



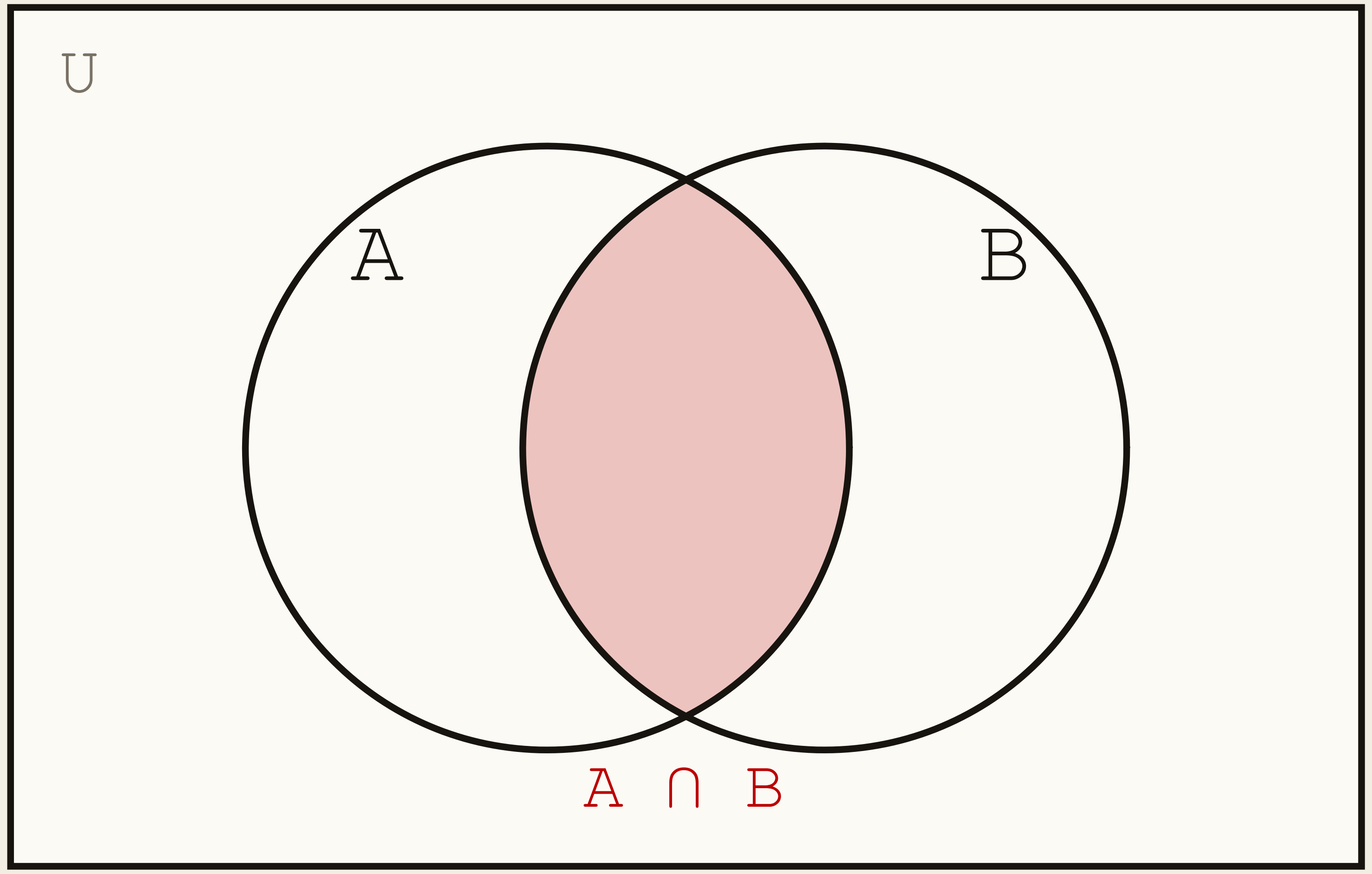
$\{1,2,3\} \cup \{1,3,5,6\}$
$= \{1,2,3,5,6\}$

Intersection

DEFINITION

The **intersection** of A and B, written $A \cap B$, contains those elements in *both* A and B.

$$A \cap B = \{x \mid x \in A \land x \in B\}$$



$$\{1,2,3\} \cap \{1,3,5,6\} = \{1,3\}$$

Every Operation Is a Connective

$A \cup B$	$x \in A \lor x \in B$	disjunction
$A \cap B$	$x \in A \land x \in B$	conjunction
$A - B$	$x \in A \land x \notin B$	conjunction with a negation
A^c	$x \notin A$	negation
$A \,\Delta\, B$	$x \in A \oplus x \in B$	exclusive or

Set theory is propositional logic pointed at membership. Everything you proved in Lecture 1 transfers.

Counting a Union

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Add the two sizes and you have counted the overlap twice. Subtract it once.

This is the two-set case of **inclusion–exclusion** . The general version arrives with counting.

CHECK IT

$A = \{1,2,3\}$ $B = \{1,3,5,6\}$

$|A| = 3,\ |B| = 4,\ |A \cap B| = 2$

$3 + 4 - 2 = 5$

$|\{1,2,3,5,6\}| = 5$ ✓

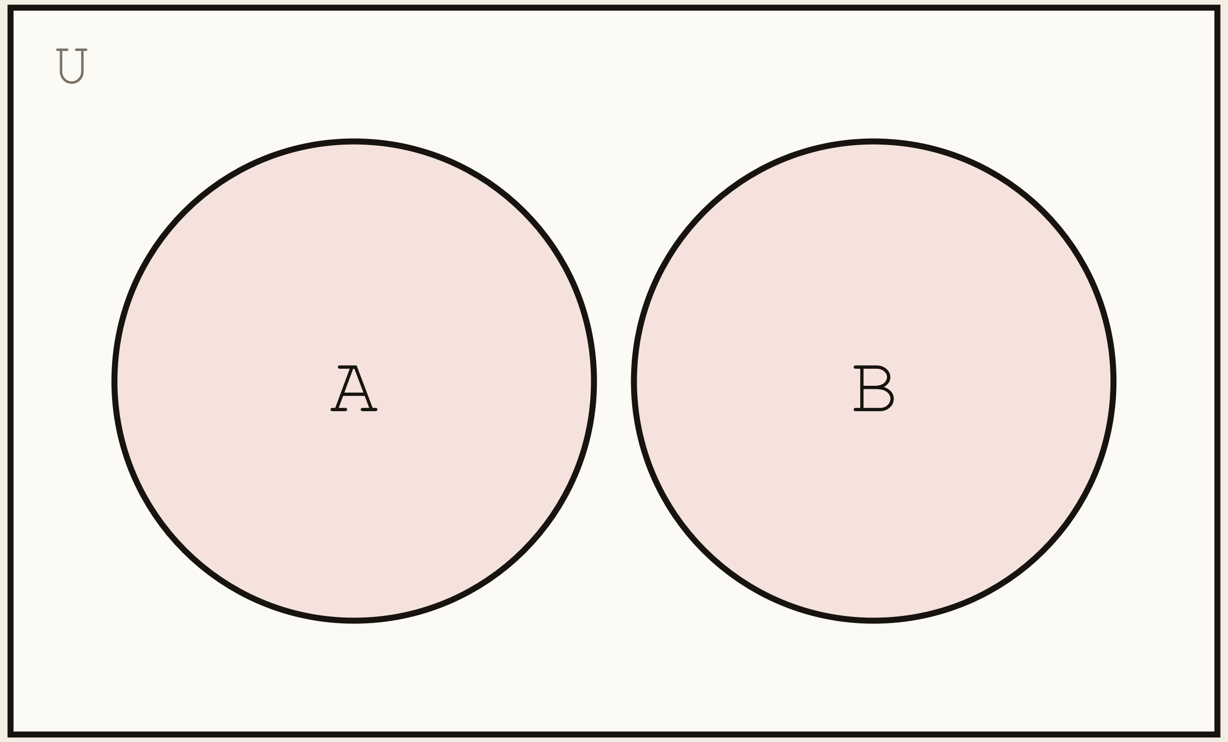
Disjoint Sets

DEFINITION

Two sets are **disjoint** if their intersection is the empty set.

$$A \cap B = \emptyset \implies |A \cup B| = |A| + |B|$$

No overlap, no correction term. Sizes simply add.



A ∩ B = ∅

The funniest Venn diagrams are the ones that don’t overlap.

Drawing the circles apart is a claim, not a decoration. It says the two sets share nothing.

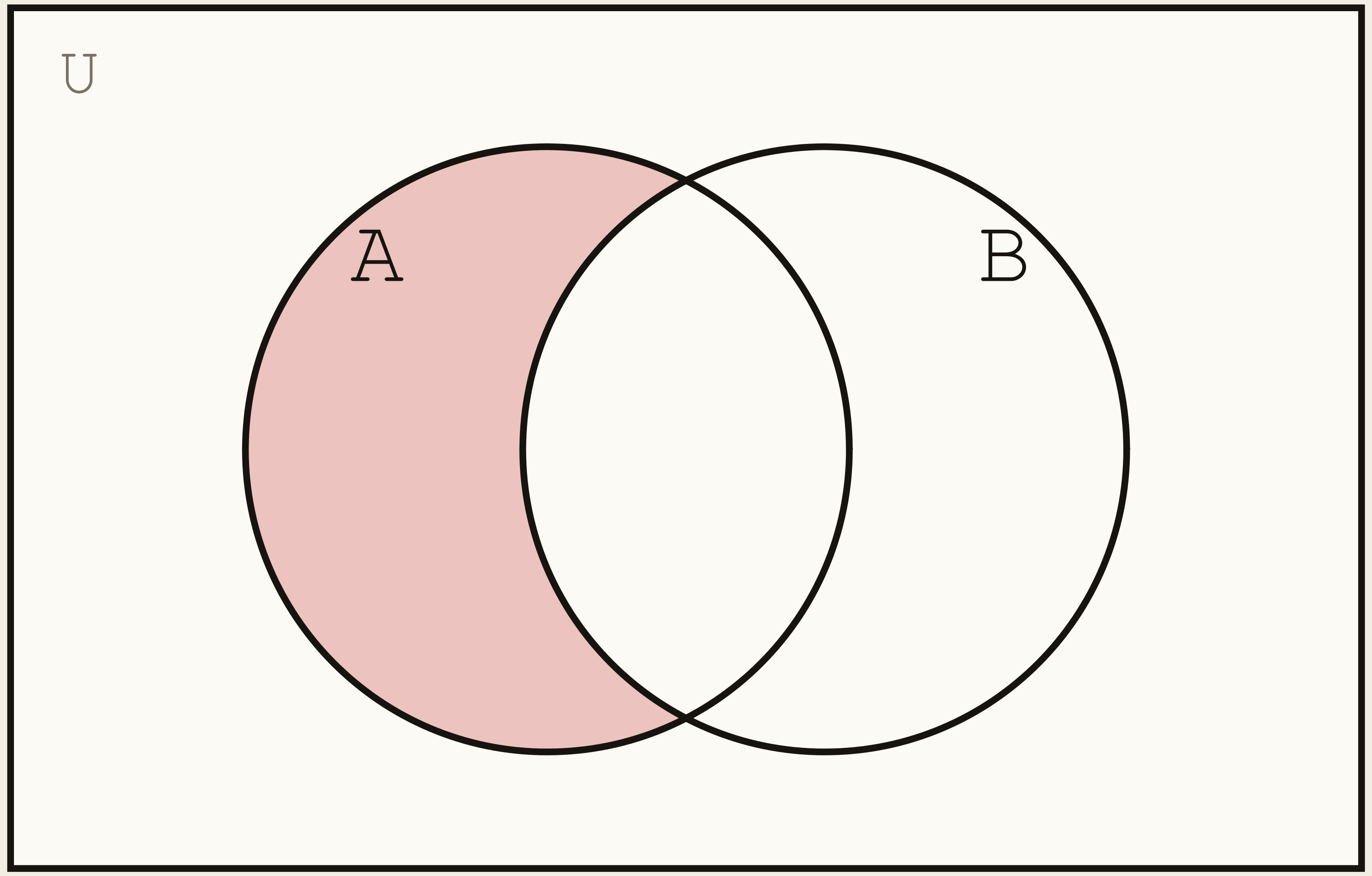
Difference

DEFINITION

The **difference** of A and B, written $A - B$, contains the elements that are in A but not in B. It is also called the **complement of B with respect to A**.

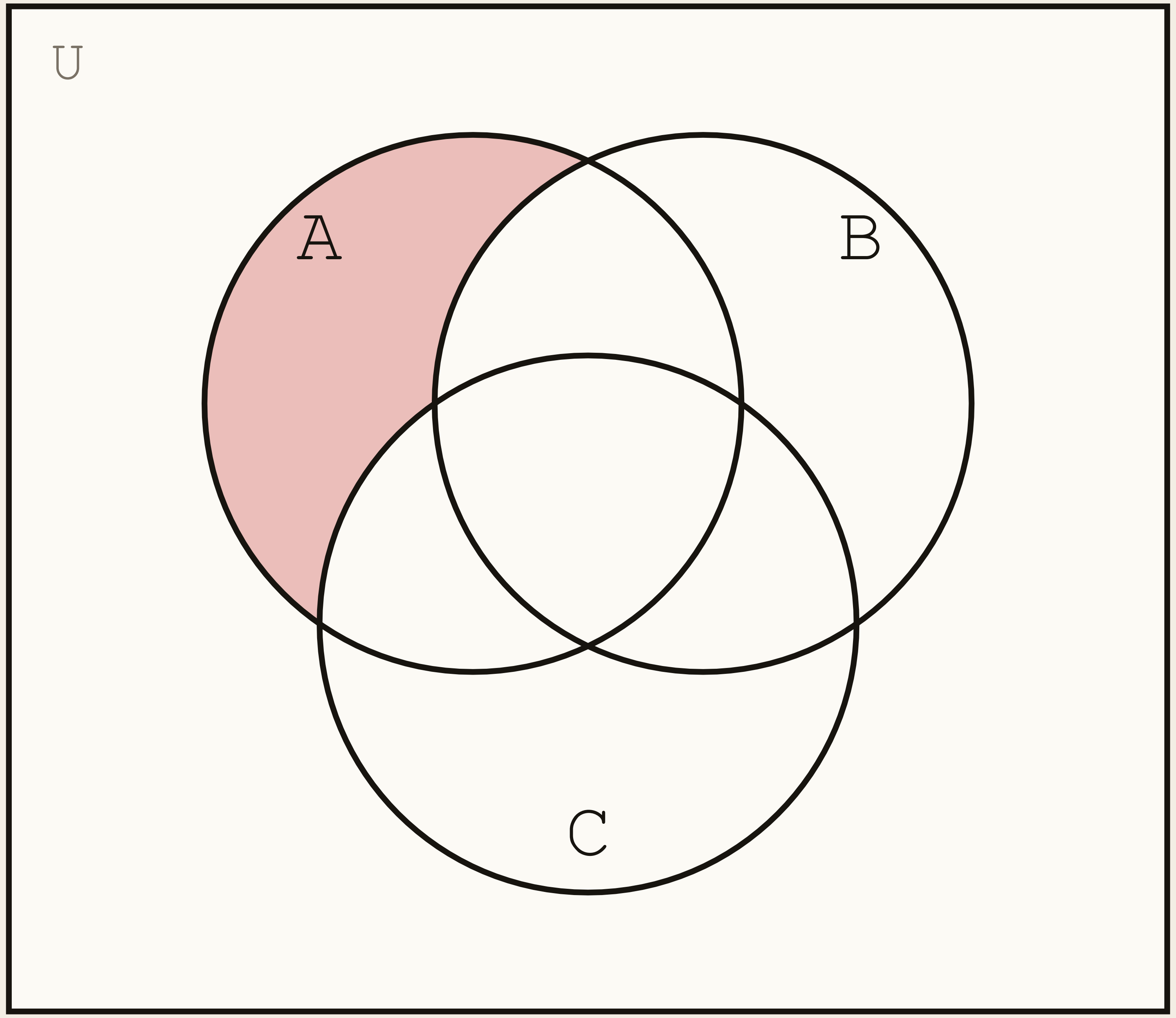
$$A - B = \{x \mid x \in A \land x \notin B\}$$

Not symmetric. Also written $A \setminus B$.



$\{1,2,3\} - \{1,3,5,6\} = \{2\}$

Reading a Region: $A - (B \cup C)$



$A - (B \cup C)$

Start with all of A. Delete anything that touches B or C.

$= A \cap (B \cup C)^c = A \cap B^c \cap C^c$

The second line is De Morgan. The picture and the algebra agree, which is the point of drawing it.

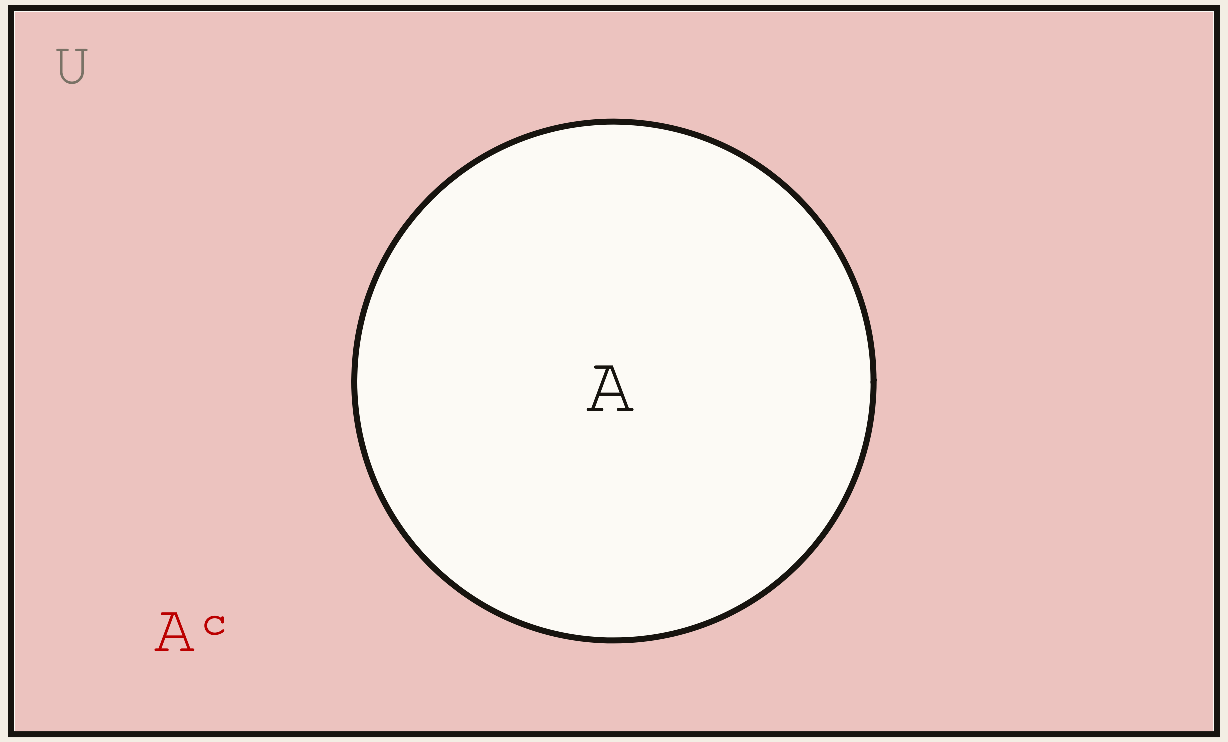
Complement

DEFINITION

Let $A \subseteq U$. The **complement** of A, written A^c, contains the elements of U that are not in A.

$$A^c = \{x \in U \mid x \notin A\}$$

Always relative to a stated U. Also written $\bar{A}$ or $U - A$.



Difference Is Just Intersection

$$A - B = A \cap B^c$$

PROOF

Take any x. Unfold both sides:

$$x \in A - B \iff x \in A \land x \notin B \iff x \in A \land x \in B^c \iff x \in A \cap B^c$$

Same elements, so the same set. ■

Worth having: it converts every difference into an intersection, which is what lets the Lecture 1 laws reach it.

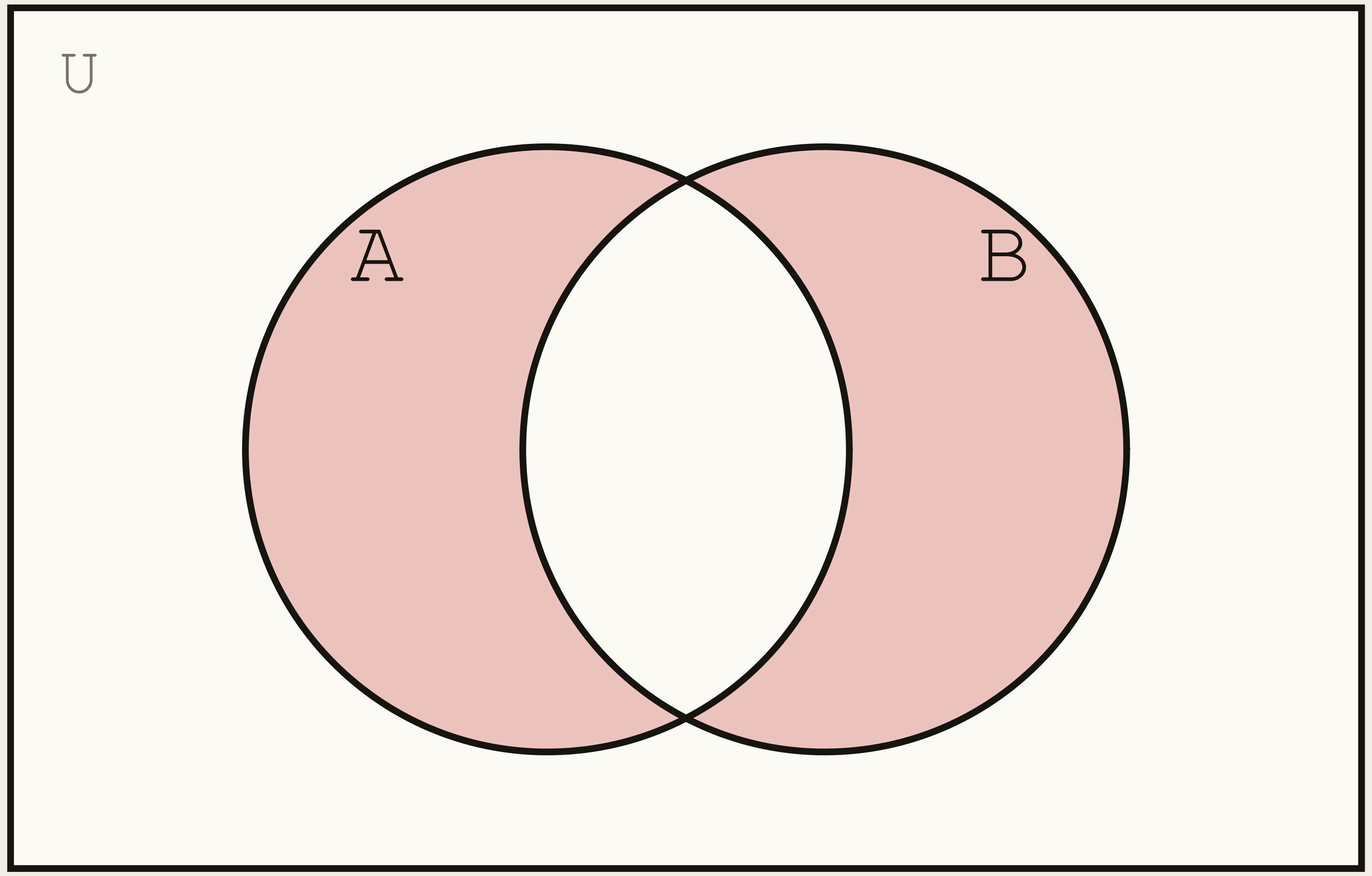
Symmetric Difference

DEFINITION

The **symmetric difference** of A and B, written $A \triangle B$, is the set of elements in either A or B but not in their intersection.

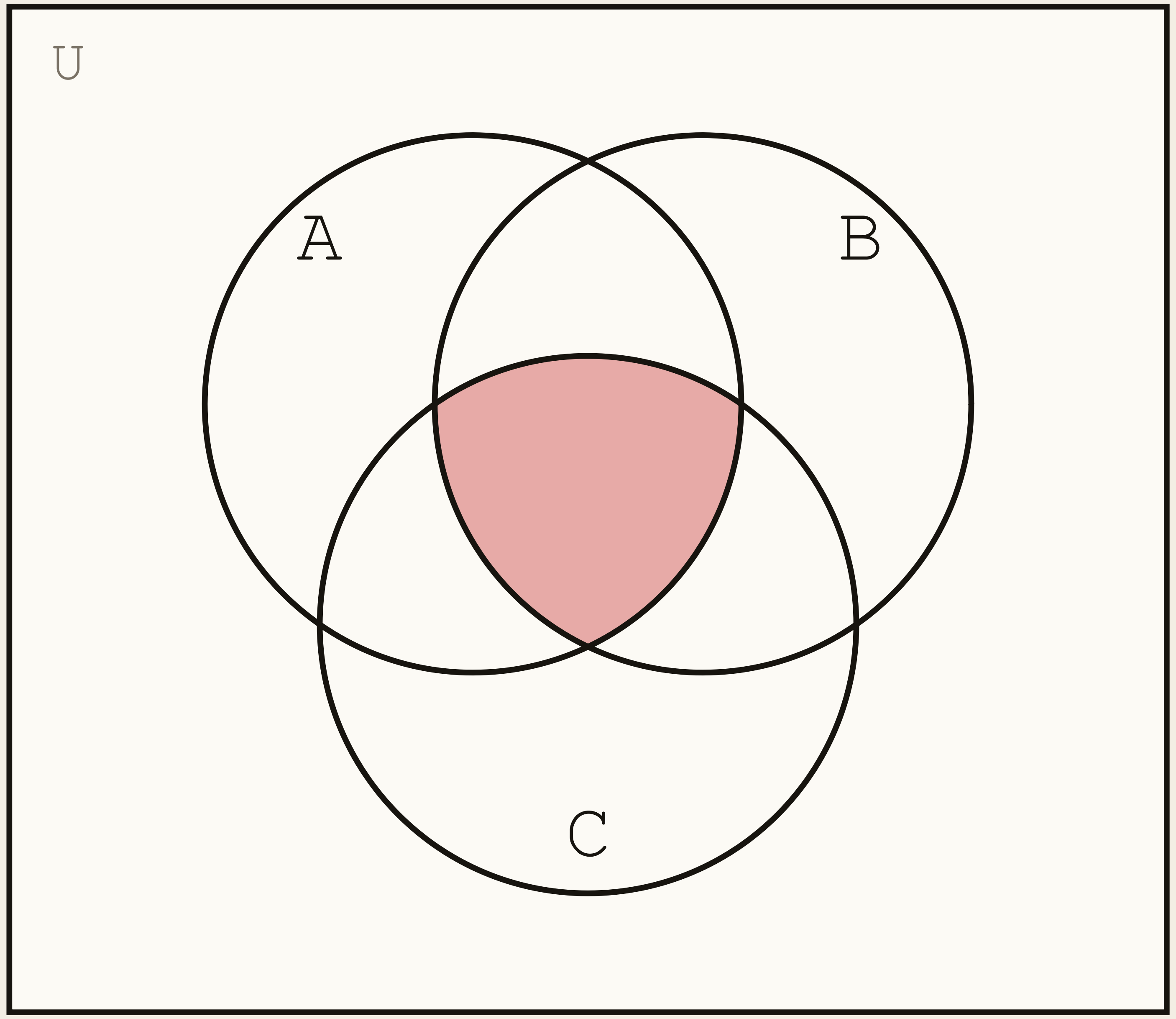
$$A \triangle B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

This one *is* symmetric: $A \triangle B = B \triangle A$. It is the $\oplus$ of Lecture 1.

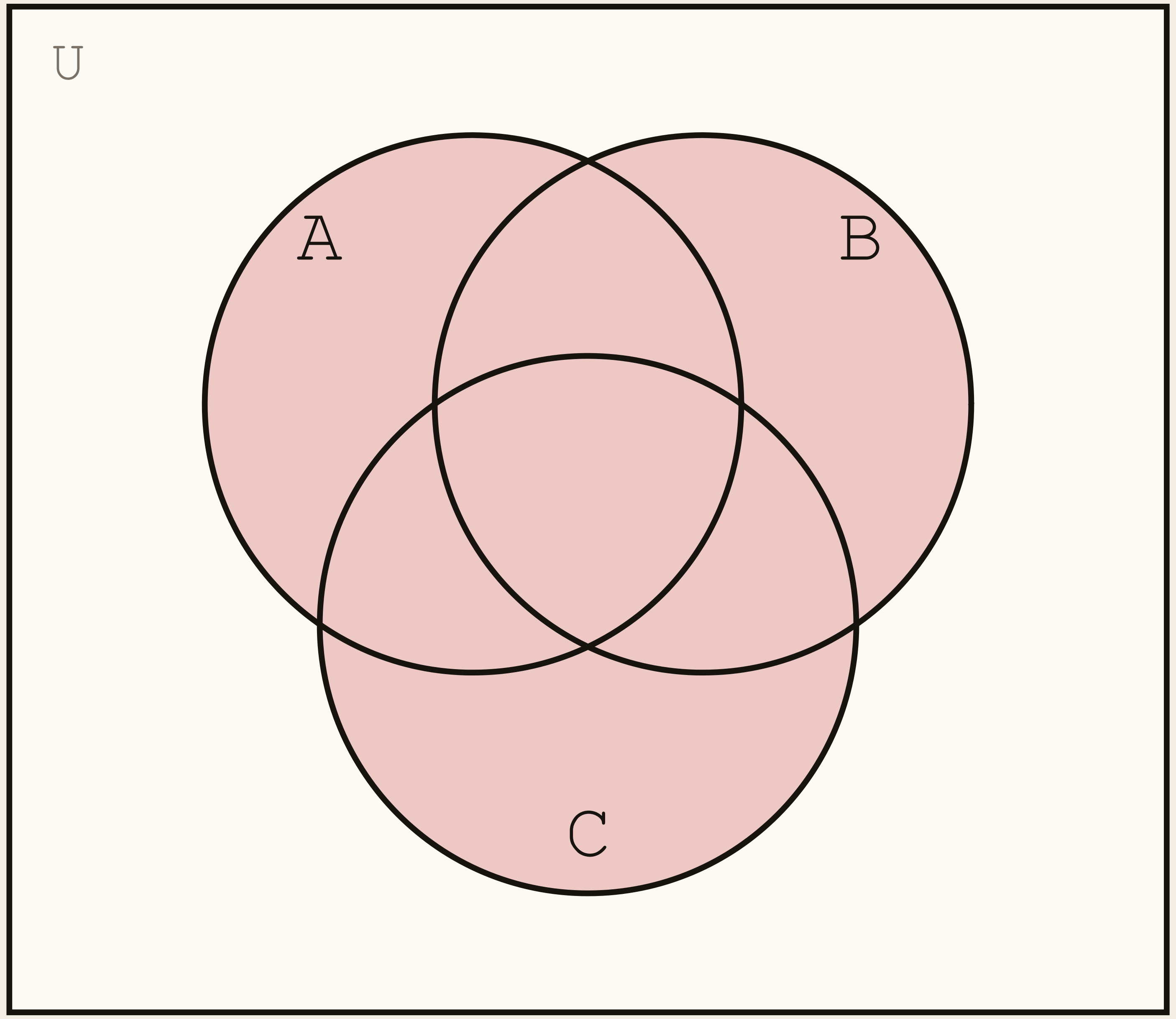


$\{1,2,3\} \triangle \{1,3,5,6\} = \{2,5,6\}$

Three Sets: The Two Extremes



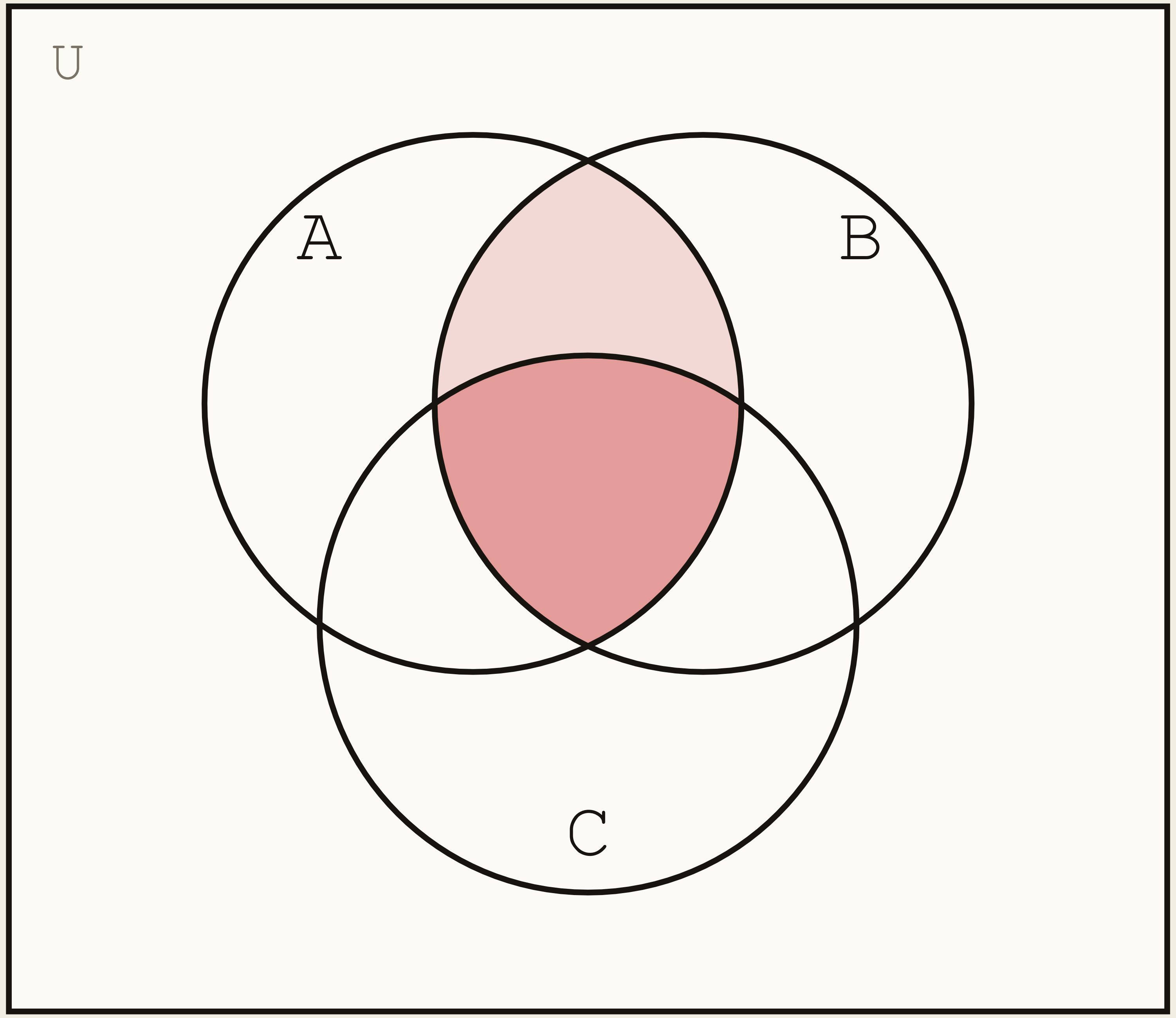
A ∩ B ∩ C



A ∪ B ∪ C

Three circles carve U into eight regions — one per in-or-out pattern. Same 2^n as always.

Splitting a Region on C

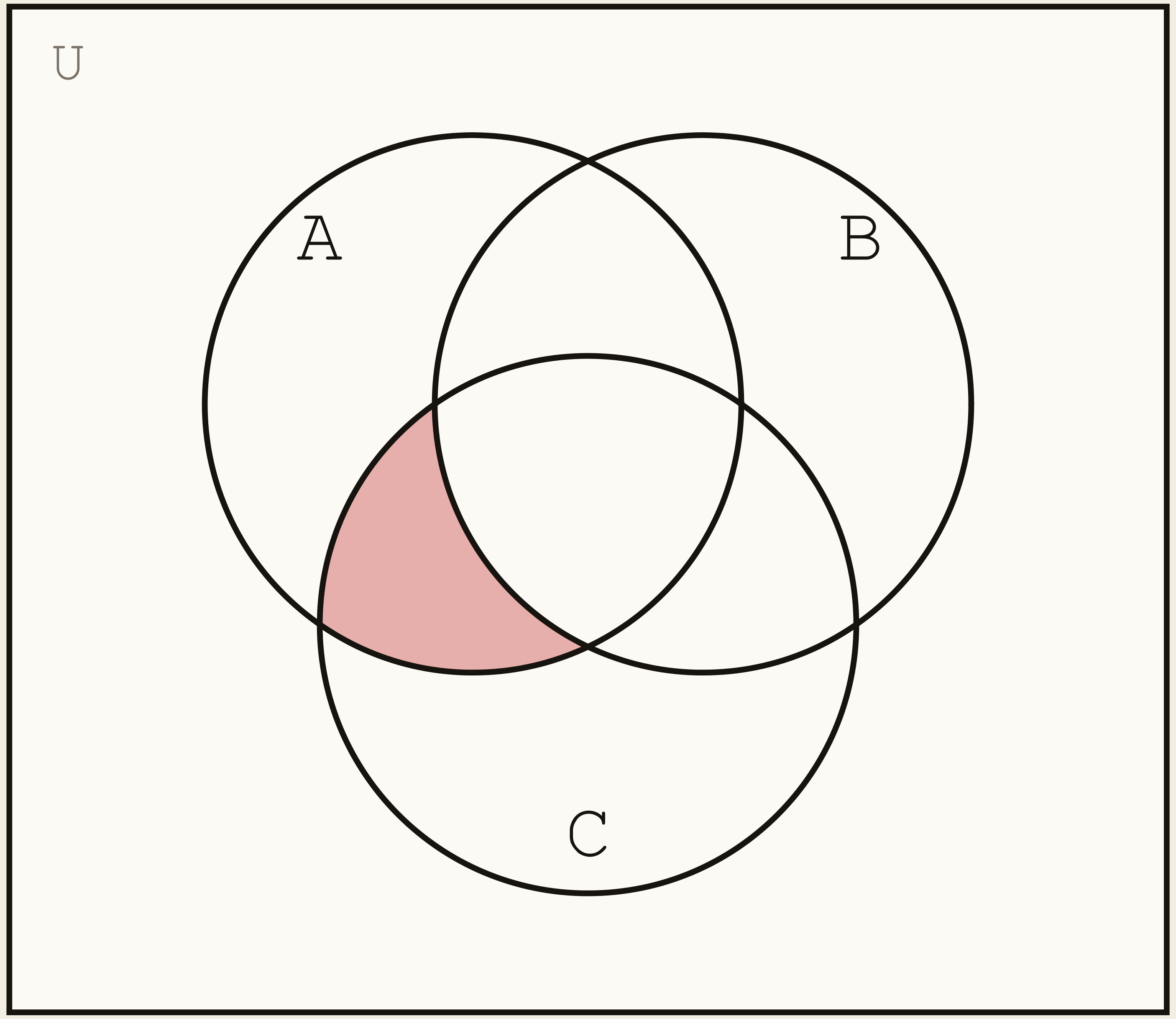


$$A \cap B = (A \cap B \cap C) \cup (A \cap B \cap C^c)$$

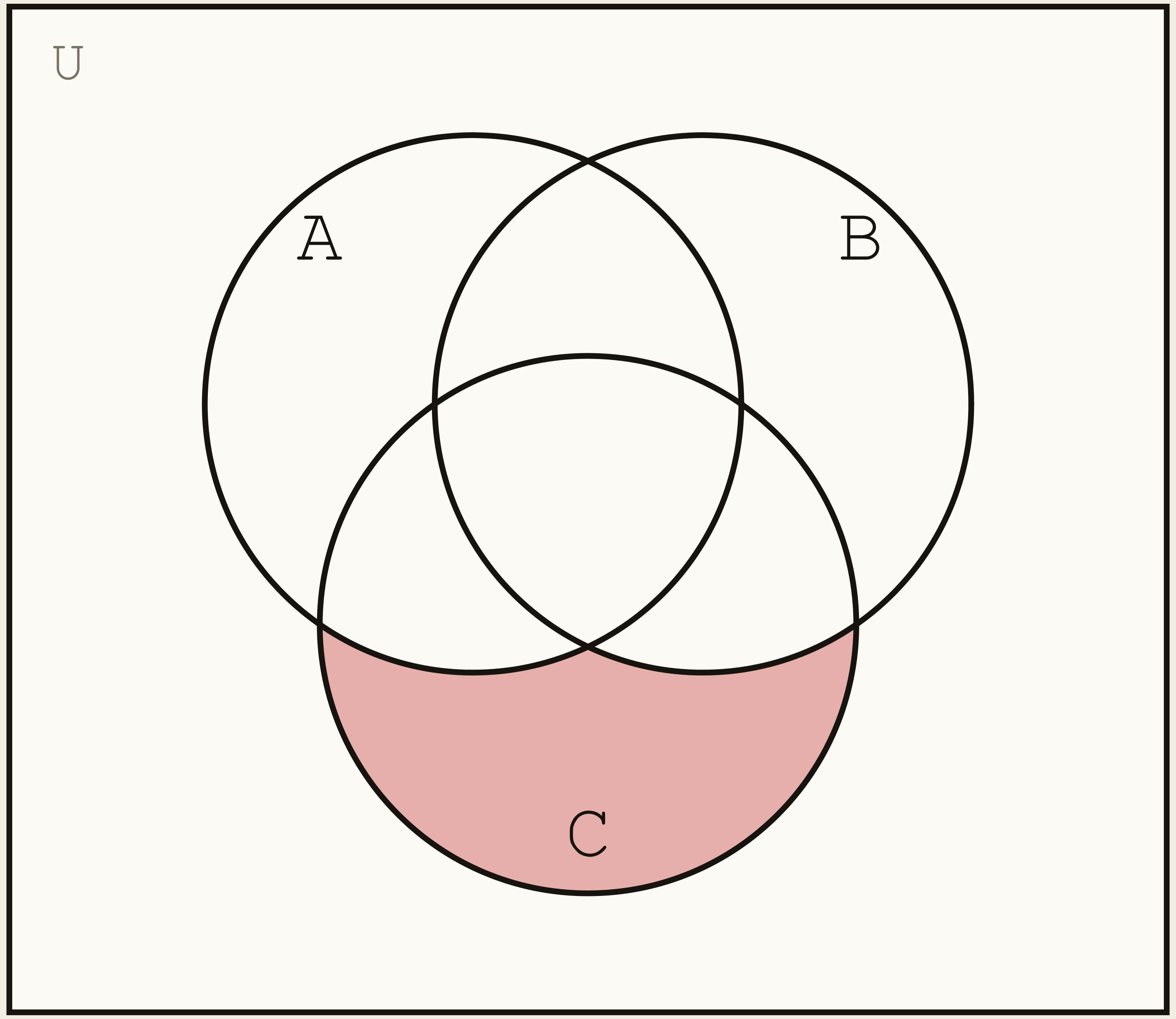
Every element of $A \cap B$ is either in C or not in C. There is no third option, and no overlap between the two cases.

Dark is the piece inside C; light is the piece outside it. Splitting on membership is how you decompose any region you can draw.

Naming a Region by Its Memberships



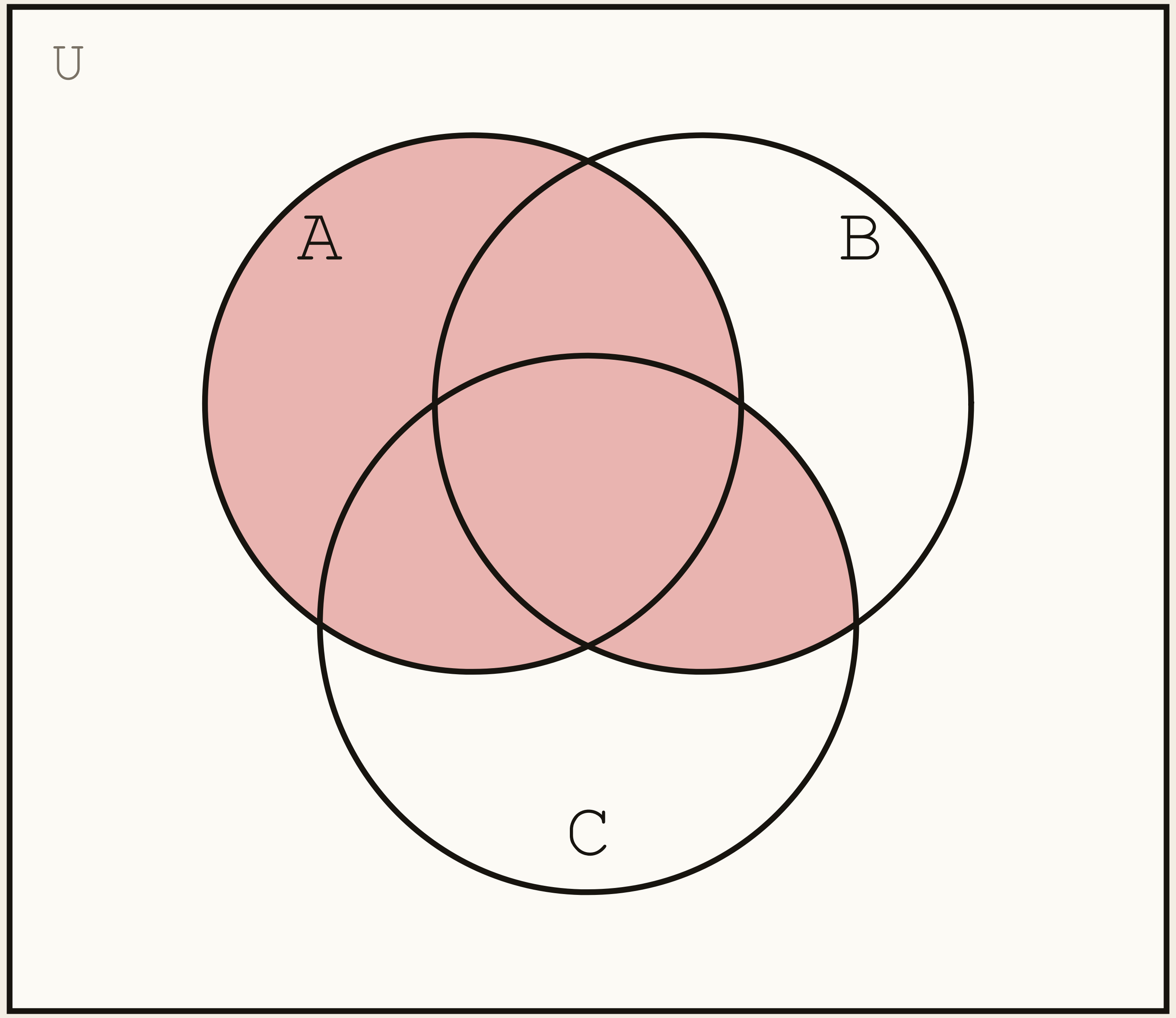
$A \cap B^c \cap C$



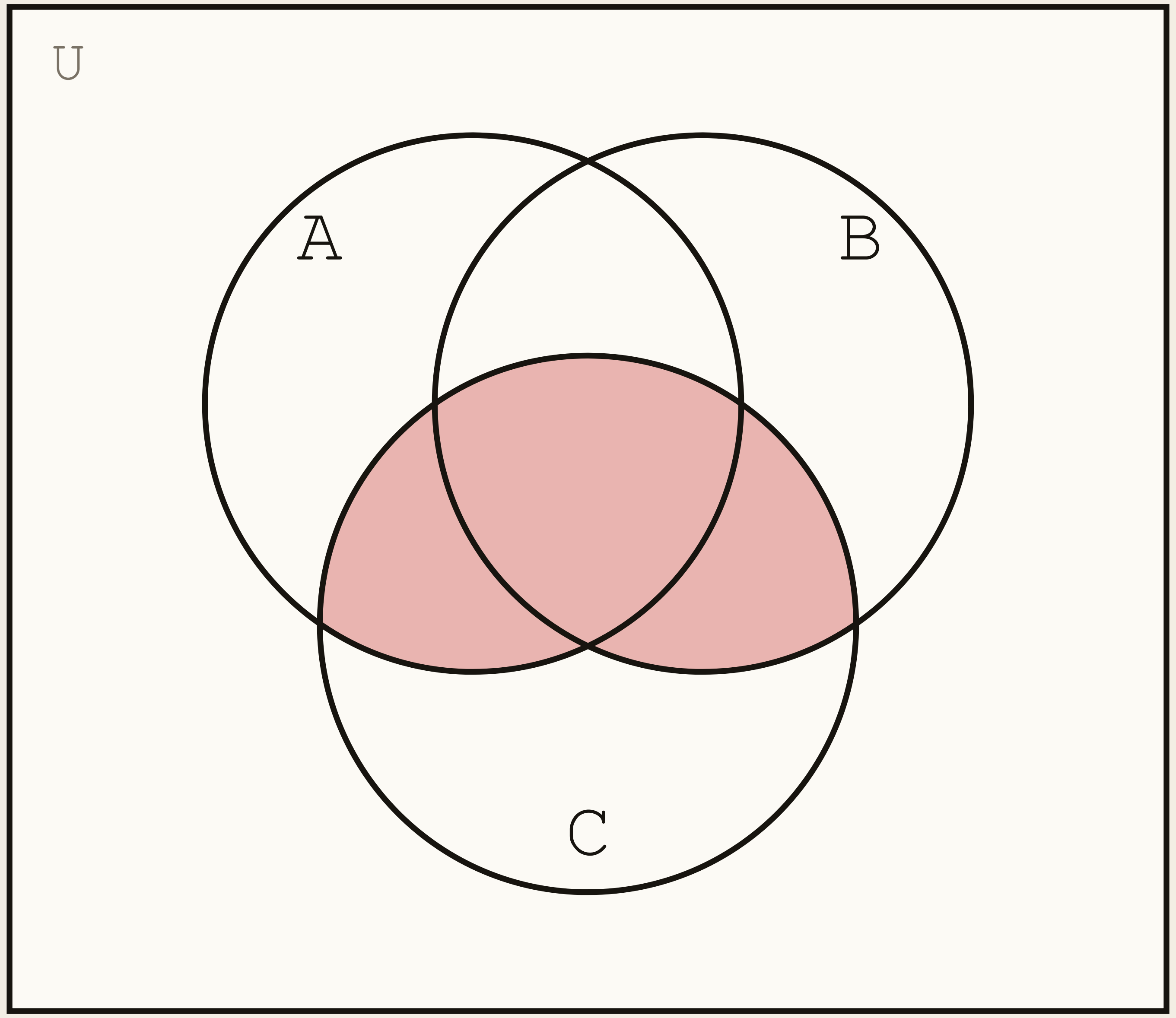
$A^c \cap B^c \cap C$

Walk the three memberships in order — in A? in B? in C? — and the name writes itself.

Parentheses Change the Set

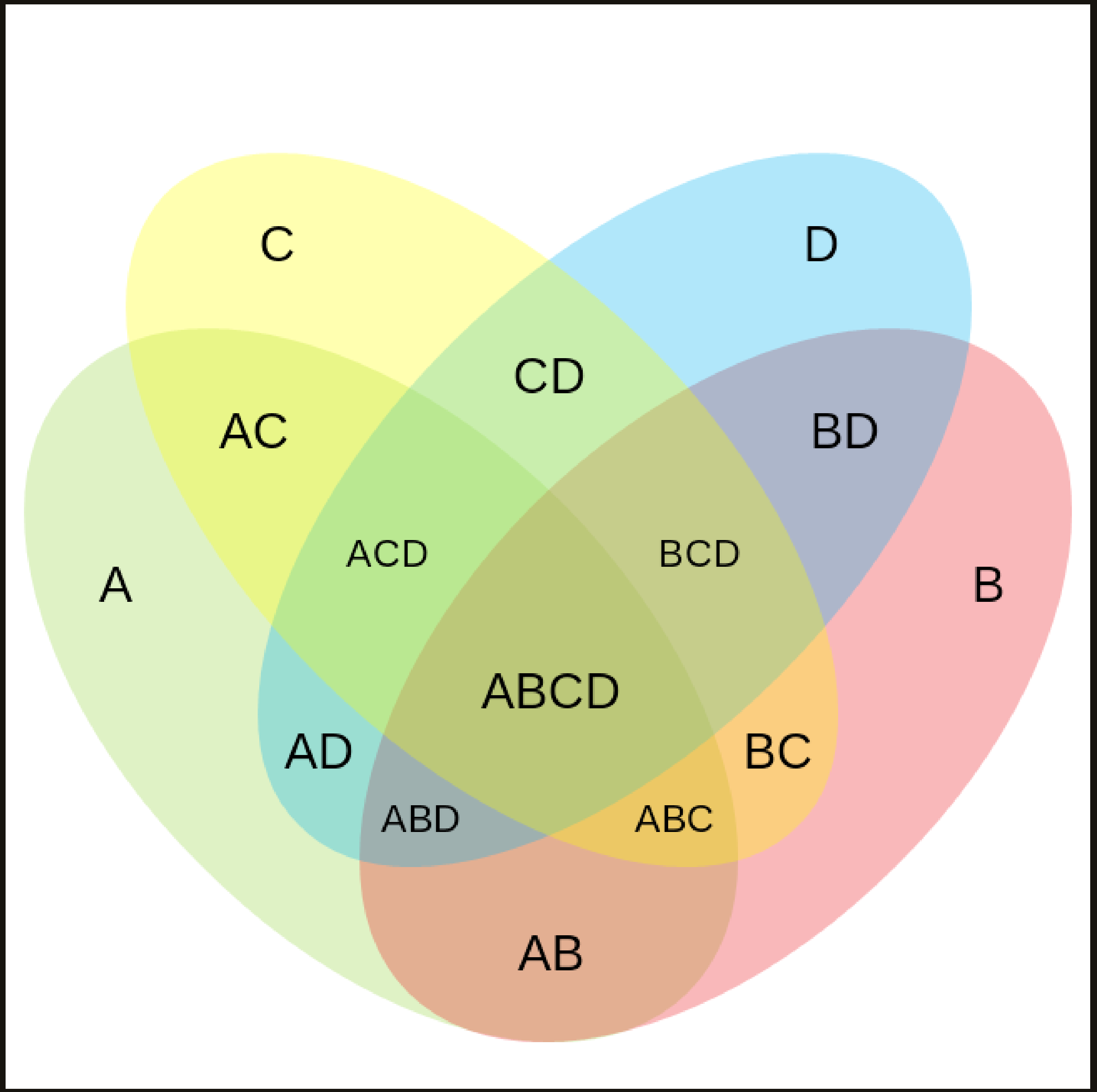


A ∪ (B ∩ C)



(A ∪ B) ∩ C

The left set contains all of A. The right set contains nothing outside C. Write the parentheses.



WHY NOBODY DRAWS FOUR

Four circles cannot make sixteen regions.

Four sets need $2^4 = 16$ regions, and four circles simply cannot produce them — it takes ellipses. Three circles is where the picture stops helping and the algebra takes over.

Set Identities

$A \subseteq A \cup B \quad B \subseteq A \cup B$

$\emptyset \cap A = \emptyset$

$A \cap B = B \cap A$

$A - B \subseteq A \quad B - A \subseteq B$

$A \subseteq B \implies A \cap B = A$

$A \subseteq B \implies A - B = \emptyset$

$A \cap B \subseteq A \quad A \cap B \subseteq B$

$\emptyset \cup A = A$

$A \cup B = B \cup A$

$A \,\Delta\, B = B \,\Delta\, A$

$A \subseteq B \implies A \cup B = B$

$A - B = A \cap B^c$

The three in scarlet all say the same thing from different angles: if A sits inside B, then B adds nothing to A and removes all of it.

Proving an Identity: Both Directions

PROOF

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$\subseteq$ LEFT INSIDE RIGHT

Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B \cup C$.

We do not know which side of the union, so split:

Case 1. $x \in B$. With $x \in A$, that gives $x \in A \cap B$.

Case 2. $x \in C$. With $x \in A$, that gives $x \in A \cap C$.

Either case puts x in the union $(A \cap B) \cup (A \cap C)$. ■

$\supseteq$ RIGHT INSIDE LEFT

Let $x \in (A \cap B) \cup (A \cap C)$.

Again a union, so again two cases:

Case 1. $x \in A \cap B$. Then $x \in A$ and $x \in B$, so $x \in B \cup C$.

Case 2. $x \in A \cap C$. Then $x \in A$ and $x \in C$, so $x \in B \cup C$.

Either case gives $x \in A$ and $x \in B \cup C$. ■

Both containments hold, so the sets are equal. Notice the rhythm: an $\cap$ in your hypothesis hands you two facts at once, while a $\cup$ hands you a case split. That is the whole craft.

De Morgan, Again

$$(A \cup B)^c = A^c \cap B^c \qquad \neg(p \lor q) \equiv \neg p \land \neg q$$

$$(A \cap B)^c = A^c \cup B^c \qquad \neg(p \land q) \equiv \neg p \lor \neg q$$

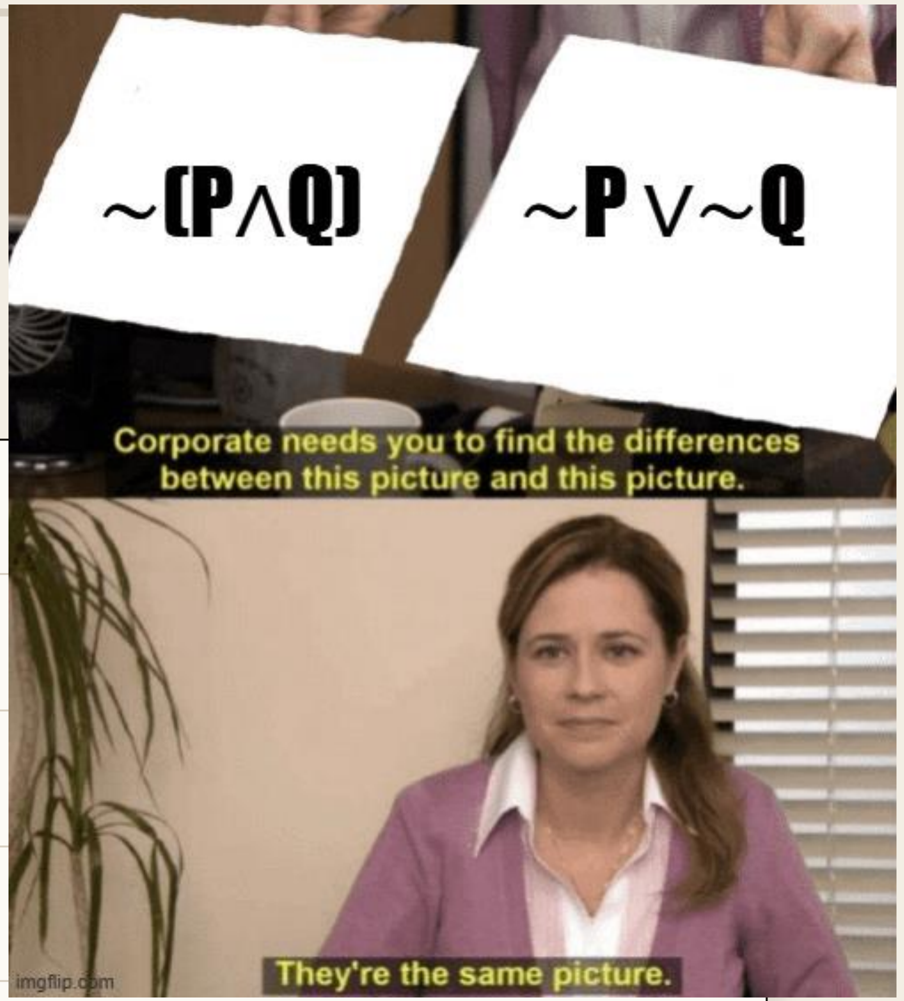
Complement is negation, union is $\lor$, intersection is $\land$. Substitute, and the set version is the propositional version. You proved this in Lecture 1.

De Morgan, Proved

PROOF

Every step below is an **equivalence**, so the chain proves both containments at once.

SHOW $(A \cup B)^c = A^c \cap B^c$	
$x \in (A \cup B)^c$	start with an arbitrary member
$\Leftrightarrow x \notin A \cup B$	definition of complement
$\Leftrightarrow \neg(x \in A \lor x \in B)$	definition of union
$\Leftrightarrow \neg(x \in A) \land \neg(x \in B)$	De Morgan for propositions — Lecture 1
$\Leftrightarrow x \in A^c \land x \in B^c$	definition of complement
$\Leftrightarrow x \in A^c \cap B^c$ ■	definition of intersection



Only the scarlet line proves anything. The other five unfold notation. That is what "set theory is logic applied to membership" means in practice — and the same chain with ∧ and ∨ swapped gives the other De Morgan law.

Membership Tables

Same identity, checked mechanically. Write 1 for "is a member," 0 for "is not" — a truth table wearing set notation.

$x \in A$	$x \in B$	$A \cup B$	$(A \cup B)^c$	A^c	B^c	$A^c \cap B^c$
1	1	1	**0**	0	0	**0**
1	0	1	**0**	0	1	**0**
0	1	1	**0**	1	0	**0**
0	0	0	**1**	1	1	**1**

The two scarlet columns agree on every row, so the identity holds. Four rows is not a sample — it is *every* possible membership pattern for two sets.

The cost: n sets need 2^n rows. Three sets is eight, four is sixteen. Past three, element chasing is faster — and it is the only method that survives infinite families.

Simplifying With the Identities

Once the identities are available you can work without elements at all. Simplify $(A \cap B) \cup (A \cap B^c)$.

ONE NAMED STEP AT A TIME

$(A \cap B) \cup (A \cap B^c)$	the expression we were handed
$= A \cap (B \cup B^c)$	distributive law — factor out A
$= A \cap U$	complement law: $B \cup B^c = U$
$= A$	identity law: $A \cap U = A$

Three substitutions, no elements chased. Every line names the law that licensed it — an unnamed step is not a proof.

Sanity check it against a diagram: every element of A is either in B or not in B. Split A on that question, glue the halves back, and you have A.

A Function Is a Set of Pairs

DEFINITION

A **function** $f : A \to B$ is a relation $f \subseteq A \times B$ such that

$$\forall x \in A\ \exists! y \in B\ ((x, y) \in f)$$

Every input has **exactly one** output. We write $f(x) = y$ for $(x, y) \in f$ — familiar notation for a set of pairs.

Nothing new — a Cartesian product from last lecture with one rule attached. A is the **domain**, B the **codomain**. The $\exists!$ can fail two ways, and both are below: *too many* outputs, or *none*.

A FUNCTION ✓

$\{ (1,a), (2,a), (3,b) \}$

Each of 1, 2, 3 appears exactly once. Two inputs *sharing* an output is fine.

NOT A FUNCTION ✗

$\{ (1,a), (1,b), (2,a), (3,b) \}$

1 appears twice — one input, two outputs. Breaks **uniqueness**.

NOT A FUNCTION ✗

$\{ (1,a), (2,a) \}$

3 never appears — nothing to return. Breaks **existence**.

The vertical line test from algebra is exactly this rule: a vertical line crossing a graph twice means one input with two outputs.

Injective — One-to-One

DEFINITION

$$\forall x_1, x_2 \in A \; (x_1 \neq x_2 \implies f(x_1) \neq f(x_2))$$

In words: distinct inputs always give distinct outputs. No two arrows land on the same target.

THE CONTRAPOSITIVE — WHAT YOU ACTUALLY USE

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

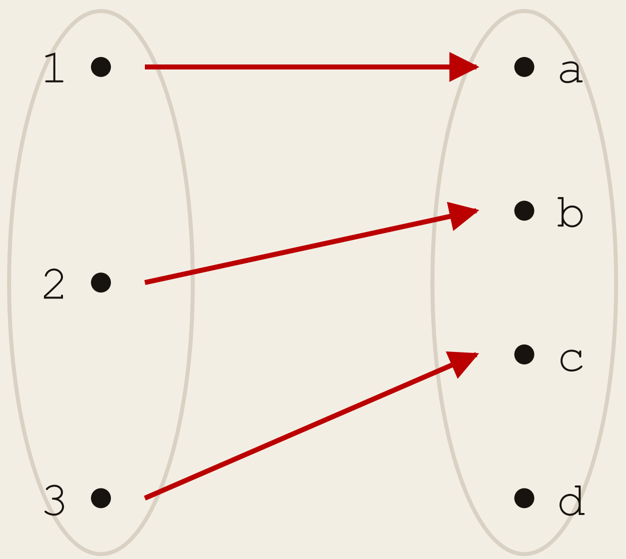
Logically identical to the definition above (Lecture 1), and far easier to work with — you get an equation to solve instead of an inequality to preserve.

$f(x) = 2x + 1$ ✓

$2x_1+1 = 2x_2+1$ forces $x_1 = x_2$.

$f(x) = x^2$ on $\mathbb{R}$ ✗

$f(2) = f(-2) = 4$, but $2 \neq -2$.



Injective, not surjective — d is missed.

Surjective — Onto

DEFINITION

$\forall y \in B\ \exists x \in A\ (f(x) = y)$

In words: every element of the codomain is hit by something. Nothing in B is left out.

HOW YOU DISCHARGE THE ∃

given y, construct $x = f^{-1}(y)$

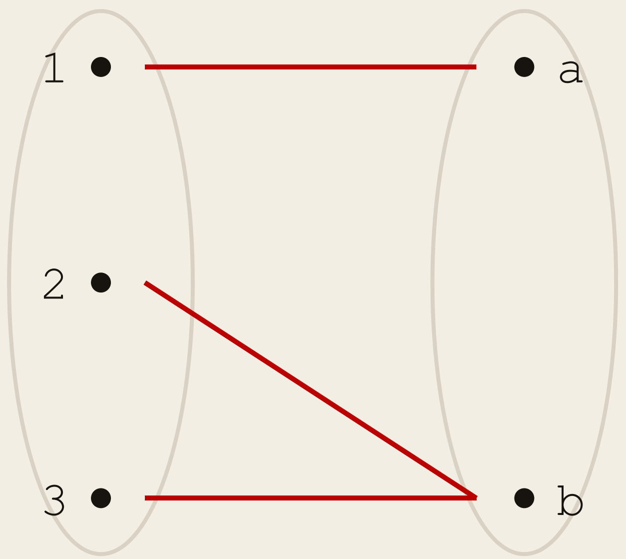
The ∃ is satisfied by *exhibiting* a witness: take an arbitrary y, solve $f(x) = y$ for x, and check it lands in A.

$2x + 1$ on $\mathbb{R} \to \mathbb{R}$ ✓

Given y, take x = (y−1)/2.

x^2 on $\mathbb{R} \to \mathbb{R}$ ✗

Nothing squares to −1.



Surjective, not injective — 2 and 3 collide.

Bijective — Both at Once

DEFINITION

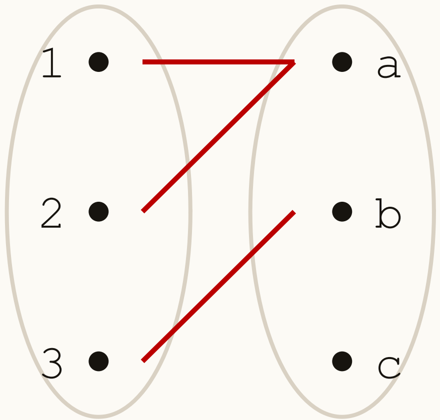
$\forall y \in B \; \exists! x \in A \; (f(x) = y)$

Read $\exists!$ as “there exists exactly one.” The $\exists$ is surjectivity; the uniqueness is injectivity.

`bijective = injective + surjective`

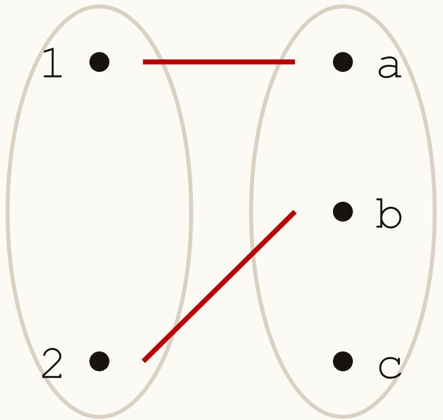
A perfect pairing: nothing on either side shared or left over. Only bijections have inverses.

NEITHER



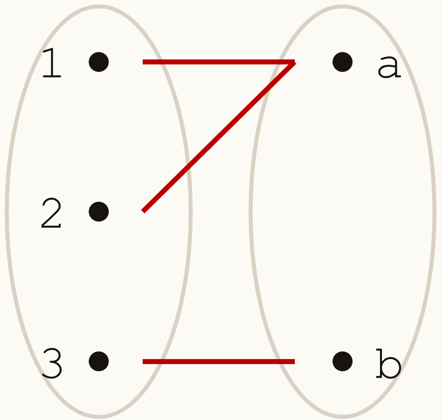
1, 2 collide · c missed

INJECTIVE ONLY



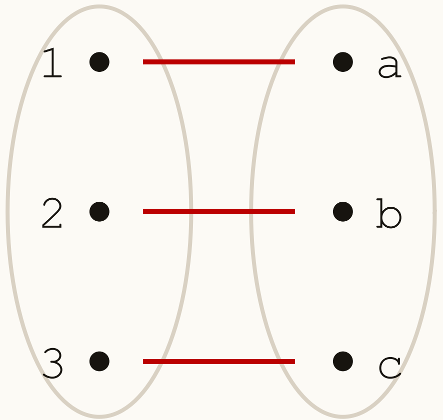
no collisions · c missed

SURJECTIVE ONLY



nothing missed · 1, 2 collide

BIJECTIVE



nothing shared · nothing missed

Read left to right: each diagram fixes one flaw. Only the last has arrows you could reverse and still have a function.

Same Formula, Four Different Answers

Every row below is $f(x) = x^2$. Only the domain and codomain change.

$f : A \to B$	injective?	surjective?	why
$\mathbb{R} \to \mathbb{R}$	no	no	±2 collide; nothing gives −1
$\mathbb{R} \to \mathbb{R}_{\geq 0}$	no	yes	shrinking B fixed onto; ±2 still collide
$\mathbb{R}_{\geq 0} \to \mathbb{R}$	yes	no	shrinking A killed the collision; −1 still missed
$\mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$	**yes**	**yes**	bijective — inverse is $\sqrt{y}$

Injective and surjective are properties of the **triple** — formula, domain, codomain — never of the formula alone.

So "is x^2 injective?" is not a complete question. If a problem does not state the domain and codomain, that is the first thing to pin down.

Practice: Which Is Which?

Both maps go $\mathbb{Z} \to \mathbb{Z}$. One is **injective but not surjective**; the other is **surjective but not injective**.

$f(n) = 2n$

n	−2	−1	0	1	2
f(n)	−4	−2	0	2	4

$g(n) = \lfloor n / 2 \rfloor$

n	0	1	2	3	4
g(n)	0	0	1	1	2

Which is which — and why?

For each negative claim, name **one** element that proves it.

Practice: The Answer

$f(n) = 2n$

Injective ✓ $2n_1 = 2n_2 \implies n_1 = n_2$

Not surjective ✗ Nothing maps to 3 — that needs $n = 3/2 \notin \mathbb{Z}$.

$g(n) = \lfloor n / 2 \rfloor$

Surjective ✓ Any m is $g(2m)$, and $2m \in \mathbb{Z}$.

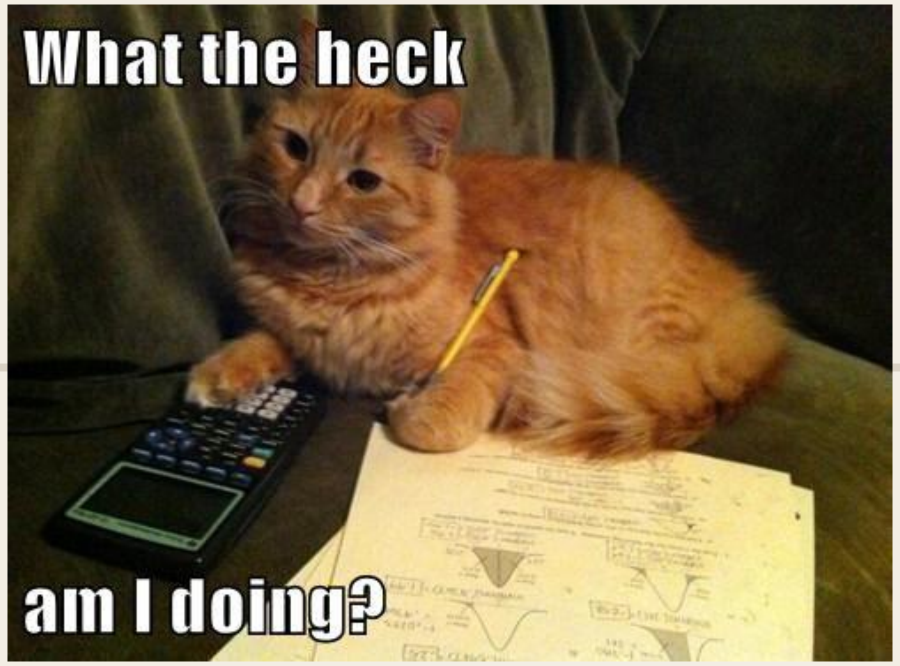
Not injective ✗ $g(4) = g(5) = 2$, but $4 \neq 5$.

$g(f(n)) = n$ but $f(g(5)) = 4$

Near-inverses, composing cleanly one way only. Neither is bijective, so neither has a true inverse.

Both negative claims were settled by **one counterexample each**. Testing more values proves nothing — a single failure is the entire proof.

Bijections Measure Size



DEFINITION

Two sets have the **same cardinality**, written $|A| = |B|$, exactly when a bijection $f : A \to B$ exists.

You can prove two sets are the same size **without counting either one** . Exhibit a pairing that misses nothing and doubles up on nothing, and you are done.

A shepherd checking sheep against pebbles never counts. One pebble per sheep, none left over: the flock is intact.

injective only

$|A| \le |B|$ — A fits inside B with room to spare.

surjective only

$|A| \ge |B|$ — A has enough to cover B , maybe too much.

bijective

$|A| = |B|$ — exactly enough, with nothing spare.

This is the *definition* of “same size,” not a consequence of it. That matters because it is the only version that survives when the sets are infinite — where counting is not available at all.

A Set the Same Size as Half of Itself

Let $E = \{0, 2, 4, 6, \ldots\}$, the even naturals. Certainly $E \subsetneq \mathbb{N}$ — every even number is natural, and $3 \in \mathbb{N}$ is not even. So E is a *strict* subset.

YET THIS PAIRING WORKS

$n \in \mathbb{N}$	0	1	2	3	4	…
$f(n) = 2n$	0	2	4	6	8	…

Injective: if $2n_1 = 2n_2$ then $n_1 = n_2$. No collisions.

Surjective: any even m is $f(m/2)$, and $m/2 \in \mathbb{N}$. Nothing missed.

$$|E| = |\mathbb{N}|$$

Throwing away half the elements changed nothing. For finite sets that is impossible — a strict subset is always strictly smaller.

Nothing above is a trick. Every step follows the definitions we just wrote down. When intuition and a verified bijection disagree, the bijection wins — and any set that pairs up with $\mathbb{N}$ this way is called **countable**.

Operations in Code

Set	Python
A ∪ B	`A | B`
A ∩ B	`A & B`
A − B	`A - B`
A Δ B	`A ^ B`
A ⊆ B	`A <= B`
A^c	`U - A`

Same operators as the bitwise ones — because a set *is* a bit string, as we saw with power sets.

There is no complement operator, for the same reason there is no universal set: the language does not know what `U` is. You have to say.

Common Mistakes

01 Treating $A - B$ as symmetric. $A - B \neq B - A$. The symmetric one is $A \triangle B$.

02 Submitting a Venn diagram as a proof. It shows you what is true; a proof says why. Prove identities by double containment.

03 Writing A^c without a universal set in sight. Complement is meaningless until U is named.

04 Using $|A \cup B| = |A| + |B|$ when the sets overlap. That only holds for disjoint sets.

Recap

01 Every set operation is a logical connective applied to membership.

02 $|A \cup B| = |A| + |B| - |A \cap B|$ — subtract the overlap you counted twice.

03 Complement needs a stated U. Difference does not: $A - B = A \cap B^c$.

04 Name a region by walking its memberships. Split a region by cases on one set.

05 The set identities are the Lecture 1 laws. Prove them by containment, by equivalence chain, or by membership table — never by drawing.

06 A bijection is a perfect pairing, and its existence *is* what $|A| = |B|$ means.

Diagrams to think with. Proofs to be sure.